Physical Chemistry

Thermodynamics

April 27, 2012

1. From definition of free Gibbs energy, $G = H - TS$, derive the following fundamental thermodynamic equation: (30 pts)

$$dG = VdP - SdT$$

2. Considering two phases of a substance are in equilibrium, derive the Clapeyron equation: (30 pts)

$$\frac{dP}{dT} = \frac{\Delta S}{\Delta V}$$

3. Pike's Peak in Colorado is approximately 4,300 m above sea level. What is the boiling point of water at the summit? What is the entropy change when water is heated at its boiling point at the summit? (40 pts)

$$P(h) = P(0)e^{-gMh/RT}$$

where $P(h)$ is the pressure at height h, $P(0)$ is the pressure at sea level, and M is the molar mass of air, 29.0 g/mol.

$\Delta_{vap}H^{o}$ for water is 40.79 kJ/mol.